# 209 Uniform Circular Motion

## 2.9.1 · 2.9.2 · 2.9.3 · 2.9.4 · 2.9.5 · 2.9.6 Circular motion

### 2.9.1 Radians and angular displacement

Angular displacement describes rotation about a centre. In radians it equals arc length divided by radius, with one complete turn equal to 2π rad. Convert angles to radians when using arc-length or

circular-motion equations. The same angle at different radii gives different arc lengths, so angular and linear displacement are not interchangeable.

$$\theta = \frac{s}{r}$$

θ: angular displacement (rad); s: arc length (m); r: radius (m).

### 2.9.2 · 2.9.3 · 2.9.4 Angular speed, linear speed and period

Angular speed is the angle in radians swept out per second; linear speed is distance travelled along the circle per second. At fixed angular speed, a larger radius gives greater linear speed. A complete uniform revolution covers 2πr and 2π rad, allowing both speeds to be found from the period. Their units differ.

$$\omega = \frac{\Delta\theta}{\Delta t}$$

$$v = r\omega$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$$

Uniform circular motion: ω is angular speed (rad $s^{-1}$); Δθ: swept angle (rad); Δt: time (s). Otherwise the ratio gives average angular velocity.

v: tangential speed (m $s^{-1}$); r: radius (m); ω: angular speed (rad $s^{-1}$).

Uniform circular motion: T: period (s); ω: rad $s^{-1}$; r: m; v: m $s^{-1}$.

### 2.9.5 · 2.9.6 Centripetal acceleration and resultant force

Uniform circular motion has constant speed but continuously changing velocity direction, requiring acceleration towards the centre and an inward resultant force. Centripetal force is not an extra kind of force: it is the radial resultant of real forces such as tension, gravity and friction. Do not draw it again as an additional force in a free-body diagram.

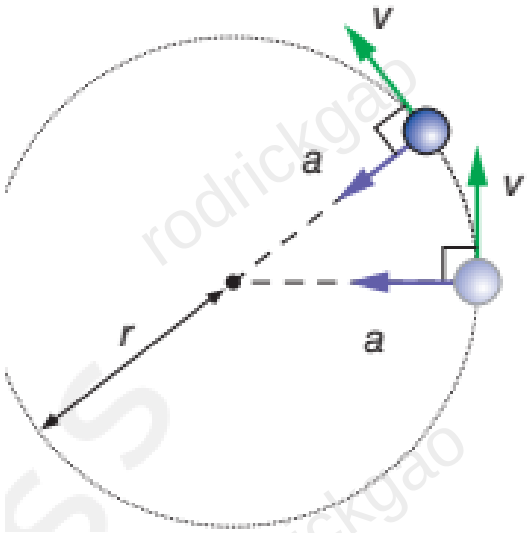
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$$a_c = \frac{v^2}{r} = r\omega^2$$

$$F_c = \frac{mv^2}{r} = mr\omega^2$$

Centripetal acceleration $a_c$ (m s$^{-2}$) points towards the centre. v: speed (m s$^{-1}$); r: radius (m); ω: rad s$^{-1}$.

$F_c$: inward resultant force (N), not an additional force; m: kg; v: m s$^{-1}$; r: m; ω: rad s$^{-1}$.




Velocity is tangential to the circle, while centripetal acceleration points towards its centre.

Textbook sources: 209 · PDF 3, 5, 6, 8, 11, 14